

Daniel A. Delgado, PhD

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WORK EXPERIENCE

University of Florida

Gainesville, FL

Enhancing Collaborator Communication in Extended Reality | XR Research & Dissertation work

August 2024 - July 2025

- Conducted extensive research and development on shared-gaze visualizations for collaborative industrial tasks in augmented reality (AR) environments.
- Designed and executed three colocated user studies (n=40+) to evaluate how visualization type, self-gaze, and environmental sound influence collaboration, social presence, and task performance.
- Developed AR prototypes integrating gaze tracking, real-time visualization, and user interface feedback to support natural interaction.

Enabling Knowledgeable Task Guidance in the Extremes (ENKIx) | User Interface Designer

June 2021 - July 2024

- Developed an augmented reality prototype for context-aware task guidance using Unity and C#/.NET.
- Implemented computer vision-based object detection with ArUco markers and designed interactive user interfaces to guide task execution.
- Collaborated with cross-functional teams to integrate networking components and align AR functionality with user research insights.

EMG-Based Prosthetic Simulation in Virtual Reality | Full-Stack Developer

June 2020 - June 2023

- Conducted a comparative study on cognitive workload and usability for training with electromyography (EMG)-based prosthetic devices using virtual reality simulations in Unity 3D versus physical devices.
- Designed and executed experimental protocols, recruited participants, and collected quantitative and qualitative usability data.

Characterizing User Comprehension in the STIR/SHAKEN Protocol | UX Researcher

June 2020 - June 2021

- Conducted a human-centered study on user interpretation of STIR/SHAKEN Caller ID authentication frameworks aimed at mitigating robocalls.
- Designed and executed surveys and interviews to evaluate consumer preferences and understanding of authentication states.

Multi-Modal Affect Detection | Machine Learning Engineer

August 2019 - August 2020

- Developed DNNs models for detecting human emotions by fusing audio, images, and video data.
- Designed and implemented feature extraction algorithms using wavelet transforms and preprocessed multimodal datasets for training supervised learning models.

EDUCATION

University of Florida

August, 2025

PhD, Computer Science

Gainesville, FL

- McKnight Doctoral Fellow, GPA: 3.6 / 4.0
- Presented cutting-edge research at international conferences.
- Competed in national collegiate road races and criteriums as part of the university cycling team.

SKILLS

User interface research & design; contextual inquiry; qualitative interviews & usability testing; quantitative surveys & data analysis; statistical analysis (R, Excel, Matlab, ANOVA, T-Tests, Regressions); survey design; computer vision; image processing; wireframing & prototyping (Figma, Sketch); stakeholder communication & presentation; participant recruitment & screening; research planning & project management; survey tools (Qualtrics, MTurk); wearables; GitHub; TensorFlow